MSP Runway Tracker

Production Hardware Assembly, Programming & Deployment Runbook

ESP32-S3-WROOM-1 | 84X WS2812B LEDS | 0.96" SSD1306 OLED | RASPBERRY PI TELEMETRY

1. System Architecture & Component Overview

This document details the exact process for flashing and commissioning the custom production PCB once your manufactured boards and components arrive.

Subsystem	Component	Pin / Bus	Functional Role
Microcontroller	ESP32-S3-WROOM-1-N16R8	Dual-Core 240MHz	16MB Flash, 8MB Octal PSRAM. Runs 30 FPS LED animation engine.
Runway LED Array	84 × WS2812B (U2–U85)	`GPIO4`	Driven via SN74LVC1T45 (U87) 5V level shifter with 330Ω series resistor.
Telemetry OLED	UCTRONICS 0.96" Yellow/Blue	`GPIO1` / `GPIO2`	Header J3 I2C bus (0x3C). Yellow: Active runway. Blue: Flight details.
UART Header J2	3-Pin Male Header	`GPIO43` / `GPIO44`	Pin 1: GND, Pin 2: TXD0 (ESP32 TX), Pin 3: RXD0 (ESP32 RX).
Main Power J1	USB-C Receptacle	5V Rail	Power input from 5V/2.5A adapter. Step-down to 3.3V via AMS1117-3.3 (U86).
Telemetry Server	Raspberry Pi 3 Model A+	Wi-Fi (wlan0)	Polls OpenSky ADS-B API and serves `http://192.168.0.213:8080/api/runway_state`.

2. Pre-Power Physical Inspection

Perform these physical verifications before applying power to prevent electrical damage:

- **Solder Inspection:** Visually inspect the castellated pads of the ESP32-S3 module under magnification. Confirm there are no solder bridges between adjacent pins, especially around `GPIO43`, `GPIO44`, `GPIO0`, `EN`, `3V3`, and `GND`.
- **OLED Header Pinout (J3):** Confirm the pin sequence of your OLED breakout board matches the PCB silkscreen. Some 0.96" SSD1306 displays use `GND-VCC-SCL-SDA`, while others use `VCC-GND-SCL-SDA`. Connecting power in reverse will destroy the display controller.
- **Multimeter Short Check:** Test continuity across the 5V power input pads (5V to GND) and across the regulated rail (3.3V to GND). Verify neither rail is shorted before connecting USB power.

> **CRITICAL: Power Supply Requirement**
> 84 WS2812B LEDs can draw substantial peak currents. **Never attempt to power the LED array from your USB-to-TTL programmer!** Always plug a dedicated 5V USB-C power supply (2.0A or greater) into receptacle **J1**.

3. Firmware Configuration & Build Environment

Your firmware is already pre-configured for your home network and live Raspberry Pi IP address:

A. Verify Network Settings in `firmware/src/config.h`

```
// Wi-Fi Credentials (2.4 GHz Band)
#define WIFI_SSID           "CenturyLinkD8F1"
#define WIFI_PASSWORD       "a7b8edf85e3f2c"

// Raspberry Pi REST Telemetry API Endpoint
#define PI_SERVER_URL       "http://192.168.0.213:8080/api/runway_state"
```

B. Build Environment: `home_wifi`

For the production board, build with `[env:home_wifi]` (not `devkit_wifi`). In this environment:

- `BOARD_MODE_WIFI=1`: Enables Wi-Fi networking and HTTP client polling.
- `BOARD_MODE_DEVKIT` is **not set**: Automatically configures pin `GPIO4` to drive all **84 LEDs** and enables hardware safety current-clamping (`MAX_LED_BRIGHTNESS = 32`).

4. Programmer Wiring & Connection (Header J2)

Connect your USB-to-TTL programmer (FTDI / CP2102) to the 3-pin Header **J2** using jumper wires:

PCB Header J2 Pin	ESP32-S3 Signal	Programmer Wire	Description
Pin 1	`GND`	**GND**	Common signal ground
Pin 2	`GPIO43` (TXD0)	**RXD**	ESP32 transmit → Programmer receive (Crossover)
Pin 3	`GPIO44` (RXD0)	**TXD**	Programmer transmit → ESP32 receive (Crossover)

> **Logic Voltage Jumper**
> Ensure the voltage jumper on your USB-to-TTL programmer is set to **3.3V logic**. Connecting 5V UART signals directly to the ESP32-S3 will cause damage.

5. Flashing the Production Hardware

1. Plug the 5V USB-C power supply into receptacle **J1** on the PCB.
2. Plug the USB-to-TTL programmer into your computer's USB port.
3. Check Windows **Device Manager** under *Ports (COM & LPT)* to confirm your COM port number (e.g. `COM3`).
4. **Enter ROM Bootloader Mode:**
   - Press and **hold SW1** (BOOT / GPIO0).
   - While holding SW1, **press and release SW2** (RESET / EN).
   - **Release SW1**. The chip is now in download mode.
5. **Upload Firmware using PowerShell:**

```
& "$HOME\.platformio\penv\Scripts\pio.exe" run -d firmware -e home_wifi -t upload --upload-port COM3
```

6. **Boot Normal Application:** Once upload completes with `[SUCCESS]`, press **SW2 (RESET)** once.

6. Post-Flash Verification & Diagnostics

A. Open Serial Monitor (115200 Baud)

```
& "$HOME\.platformio\penv\Scripts\pio.exe" device monitor -d firmware -e home_wifi -b 115200 --port COM3
```

B. Expected Boot Sequence

```
[Init] MSP Runway LED Tracker (Home Wi-Fi Mode)
[Init] OLED SSD1306 initialized (128x64 at 0x3C)
[Init] FastLED initialized: 84 WS2812B LEDs on GPIO4 (Max Brightness: 32)
[WiFi] Connecting to SSID: CenturyLinkD8F1...
[WiFi] Connected! IP Address: 192.168.0.xxx
[HTTP] Polling telemetry from http://192.168.0.213:8080/api/runway_state
[HTTP] 200 OK - Active ops: 1
```

C. Hardware Subsystem Checklist

- **OLED Display:** Displays live METAR weather and runway roles in idle periods; displays live flight details (callsign, airline, airframe, origin/destination) during flight events.
- **Runway 12R–30L (LEDs 0–25, 26 count):** Landing touchdown rollout comets start at threshold; takeoff roll accelerates in departure direction.
- **Runway 30R–12L (LEDs 26–45, 20 count):** Parallel approach and rollout tracking.
- **Runway 22–4 (LEDs 46–63, 18 count):** Crosswind runway properly passes through intersection points.
- **Runway 35–17 (LEDs 64–83, 20 count):** North-South crosswind runway animations render smoothly.

7. Troubleshooting Quick Reference

Symptom	Probable Cause	Corrective Action
`Failed to connect to ESP32-S3`	Not in bootloader mode or TX/RX swapped	Swap TX/RX on J2. Hold SW1, tap SW2, release SW1, then retry flash.
`Reason: 201 - NO_AP_FOUND`	SSID set to 5 GHz band	Ensure SSID is set to 2.4 GHz (`CenturyLinkD8F1`).
OLED display stays dark	I2C address or pin mismatch	Check Header J3 soldering. Confirm OLED pin order matches PCB silkscreen.
LEDs flicker or freeze	Voltage sag / weak power supply	Ensure USB-C supply delivers ≥ 2.0A at 5.0V. Check level shifter U87.